

Time-series Deep Learning for Time Series Forecasting

Giacomo Ziffer

giacomo.ziffer@polimi.it

PhD student @ Politecnico di Milano

Founder & CEO @ motus ml



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Supervised ML for Time Series

- For **four decades**, the mantra in applied forecasting has been **‘simpler methods prevail’**, influenced largely by the results of the Makridakis series forecasting competitions.
- Until the most **recent** M-competitions, where every top solution relied on **Machine Learning**.
- These results highlights **Machine Learning** as the **future** of time-series forecasting.

Time for ML and DL methods!

Table 1: Main performance results for TimeGPT with zero-shot inference and benchmark models measured with rMAE and rRMSE, lower scores are better. The best model for each frequency and metric is highlighted in bold, the second best underlined, and the third best underlined with a dotted line.

	Monthly		Weekly		Daily		Hourly	
	rMAE	rRMSE	rMAE	rRMSE	rMAE	rRMSE	rMAE	rRMSE
ZeroModel	2.045	1.568	6.075	6.075	2.989	2.395	10.255	8.183
HistoricAverage	1.349	1.106	4.188	4.188	2.509	2.057	2.216	1.964
SeasonalNaive	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Theta	0.839	0.764	1.061	1.061	0.841	0.811	1.163	1.175
DOTheta	0.799	0.734	1.056	1.056	0.837	0.806	1.157	1.169
ETS	0.942	0.960	1.079	1.079	0.944	0.970	0.998	1.009
CES	1.024	0.946	1.002	1.002	0.919	0.899	0.878	0.896
ADIDA	0.852	0.769	1.364	1.364	0.908	0.868	2.307	2.207
IMAPA	0.852	0.769	1.364	1.364	0.908	0.868	2.307	2.207
CrostonClassic	0.989	0.857	1.805	1.805	0.995	0.933	2.157	2.043
LGBM	1.050	0.913	0.993	0.993	2.506	2.054	0.733	0.709
LSTM	0.836	0.778	1.002	1.002	0.852	0.832	0.974	0.955
DeepAR	0.988	0.878	0.987	0.987	0.853	0.826	1.028	1.028
TFT	0.752	0.700	0.954	0.954	0.817	0.791	1.120	1.112
NHITS	<u>0.738</u>	<u>0.694</u>	<u>0.883</u>	<u>0.883</u>	0.788	0.771	<u>0.829</u>	<u>0.860</u>
TimeGPT	0.727	0.685	0.878	0.878	<u>0.804</u>	<u>0.780</u>	<u>0.852</u>	<u>0.878</u>

[src: <https://arxiv.org/pdf/2310.03589.pdf>]

Deep Learning for Time Series Forecasting

Challenges faced by standard existing models

Most methods are designed to forecasting individual series or small groups. New set of problems have emerged:

- **Forecasting many** individual or grouped **time series**.
- Trying to learn a global model facing the difficulty of **dealing with scale of different time-series** that would otherwise be related.
- Many classical models cannot **account for exogenous or multivariate inputs**.
- **Cold start problem** for new items to be included in the forecast.

Goals for novel models

- Ability to **learn** and **generalize from similar series** provides us with the ability to learn more complex models without overfitting.
- Ability to **estimate the probability distribution** of a time series' future given its past, i.e., **Probabilistic Forecasting**.

One of the first solution: DeepAR by

- DeepAR is a forecasting model based on **Autoregressive RNNs**, which learns a **global** model from historical data **of all time series** in all datasets.
- DeepAR is the **first** successful model **to combine Deep Learning** with traditional **Probabilistic Forecasting**.

DeepAR Advantages

- **Multiple time-series support:** The model is trained on multiple time-series, learning global characteristics that further enhance forecasting accuracy.
- **Extra covariates:** DeepAR allows extra features (covariates). For instance, if your task is temperature forecasting, you can include humidity-level, air-pressure etc.
- **Probabilistic output:** Instead of making a single prediction, the model leverages Monte Carlo samples to output prediction intervals.
- **“Cold” forecasting:** By learning from thousands of time-series that potentially share a few similarities, DeepAR can provide forecasts for time-series that have little or no history at all.

LSTMs in DeepAR

DeepAR uses LSTM networks to create probabilistic outputs.

LSTMs are used in numerous time-series forecasting model architectures, such as:

- Multi-stacked LSTMs
- LSTMs with CNNs
- LSTMs in encoder-decoder topology
- Etc.



DeepAR under the hood

Contrary to the previous models, DeepAR uses LSTMs a bit differently:

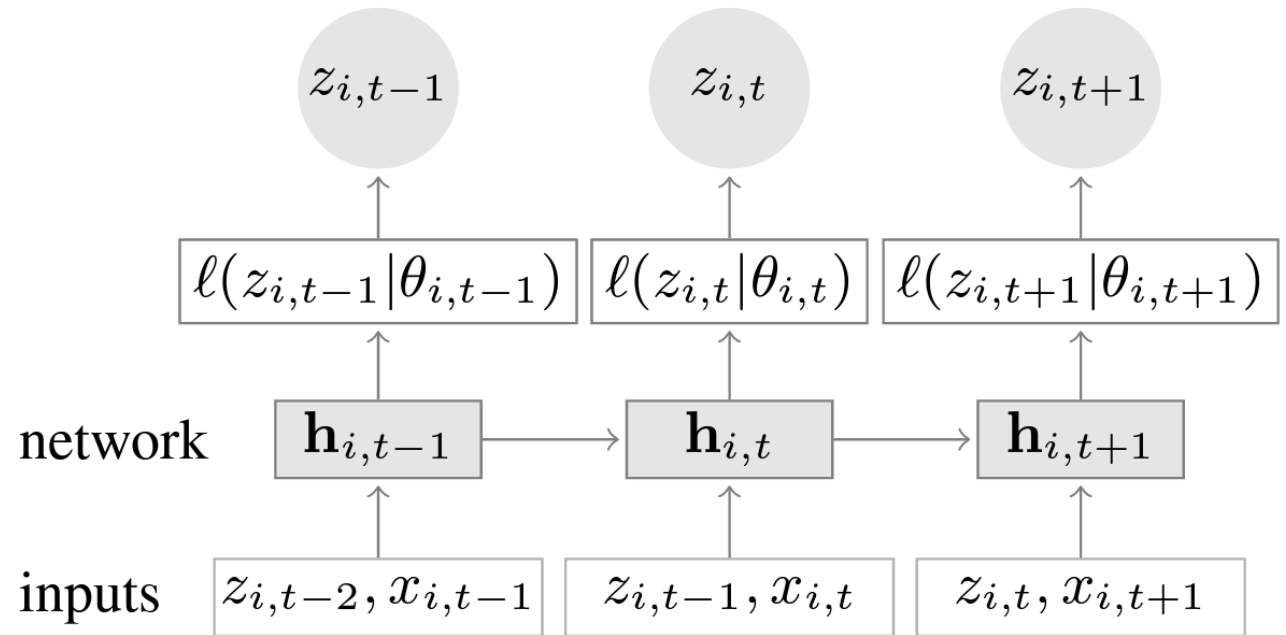
Instead of using LSTMs to calculate predictions directly,
DeepAR leverages LSTMs to predict all parameters of the probability distribution for the next time point.

For instance, DeepAR estimates the $\theta = (\mu, \sigma)$ parameters (mean and standard deviation) of the Gaussian function.

The intuition behind DeepAR training

DeepAR tries to answer to this question (assuming a Gaussian likelihood for real-valued data):

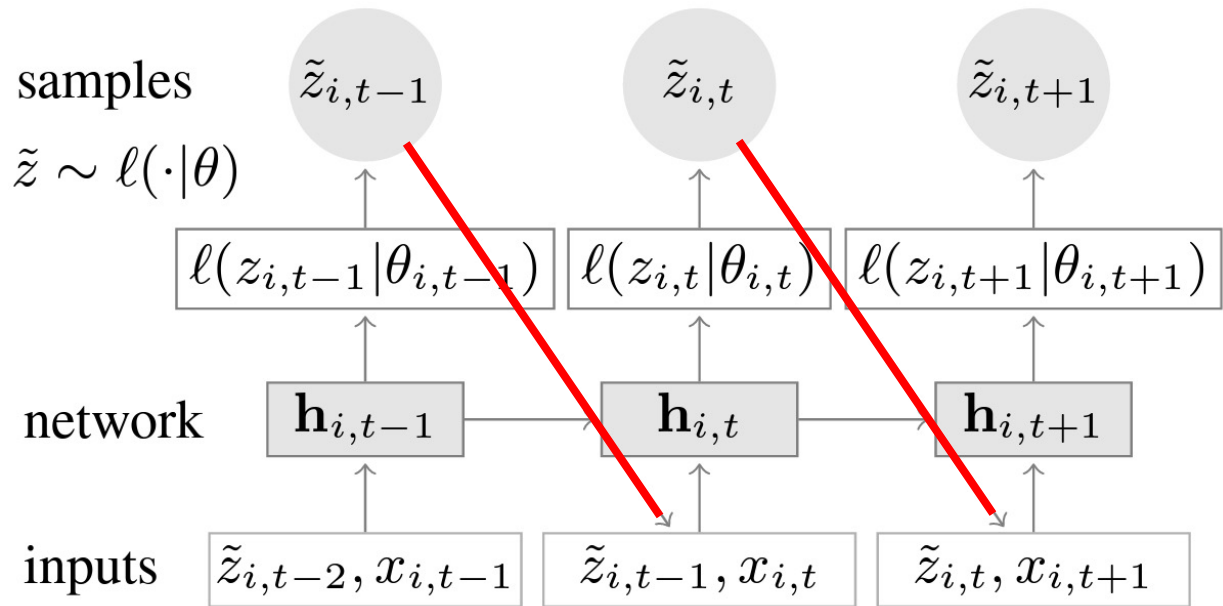
What are the best parameters μ and σ that construct a gaussian distribution which outputs predictions as close to the target variable $z_{i,t-1}$ as possible?



[src: <https://arxiv.org/pdf/1704.04110.pdf>]

DeepAR inference

W.r.t. training, one thing changes: during inference, at each inference step t , it uses the predicted variable $\tilde{z}_{i,t-1}$ that was sampled in the previous time step $t-1$ to calculate the new prediction $\tilde{z}_{i,t}$.

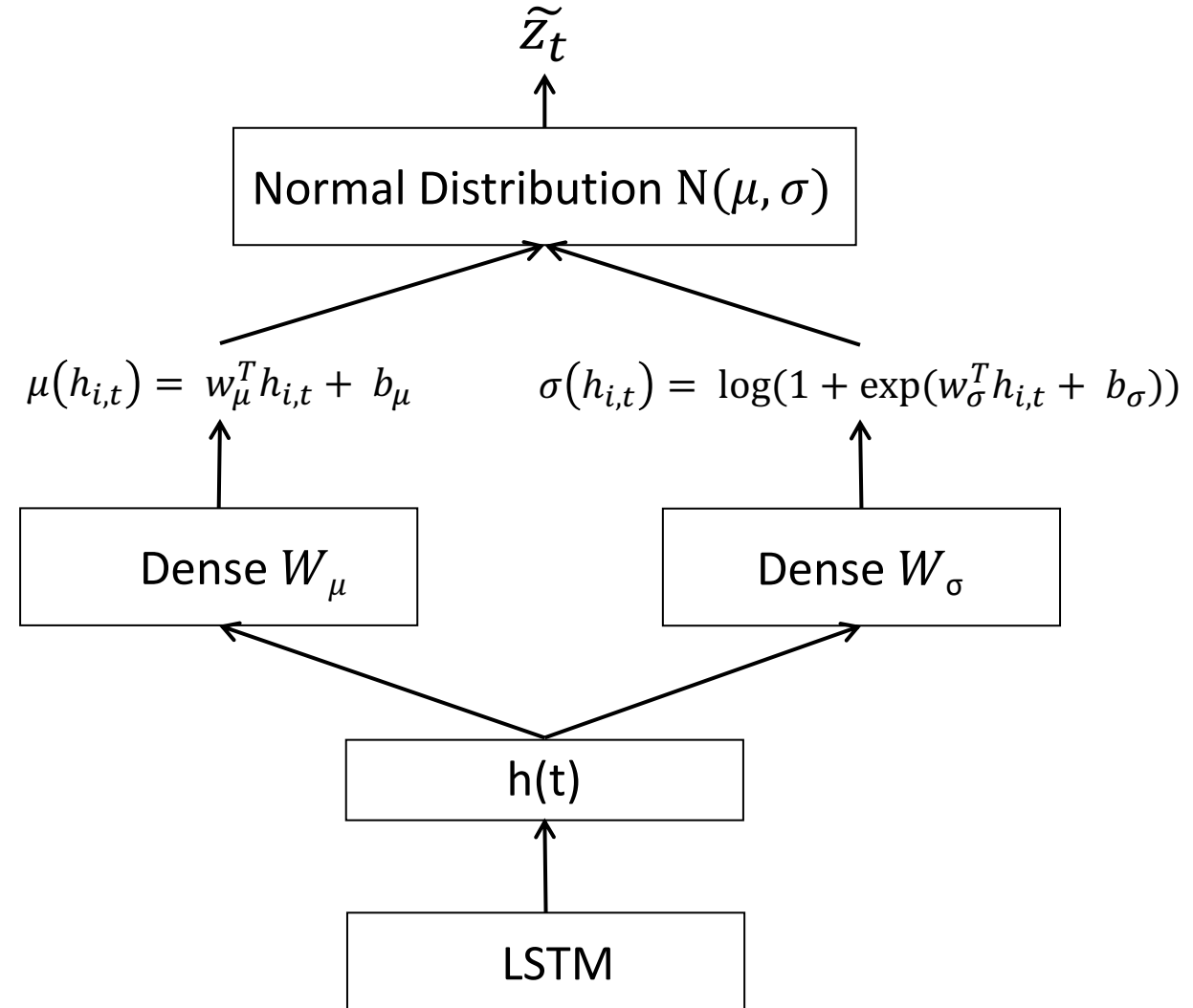


[src: <https://arxiv.org/pdf/1704.04110.pdf>]

Parameter estimation

DeepAR uses **2 Dense layers** **derive μ , σ parameters** based on the model's input.

In other words, DeepAR computes μ and σ indirectly through $h_{i,t}$, W_μ and W_σ . This is done to make their calculation possible through backpropagation.



DeepAR in the Time-Series Landscape

- **DeepAR** in various tests **outperformed traditional statistical methods** such as ARIMA.
- Also, the great advantage of DeepAR over those models is that **it does not require extra feature preprocessing** (e.g., making the time-series stationary first).
- It is **prevalent in production**: It is part of Amazon's GluonTS toolkit for time-series forecasting and it can be trained on Amazon SageMaker [1].

[1]: <https://docs.aws.amazon.com/sagemaker/latest/dg/deepar.html>

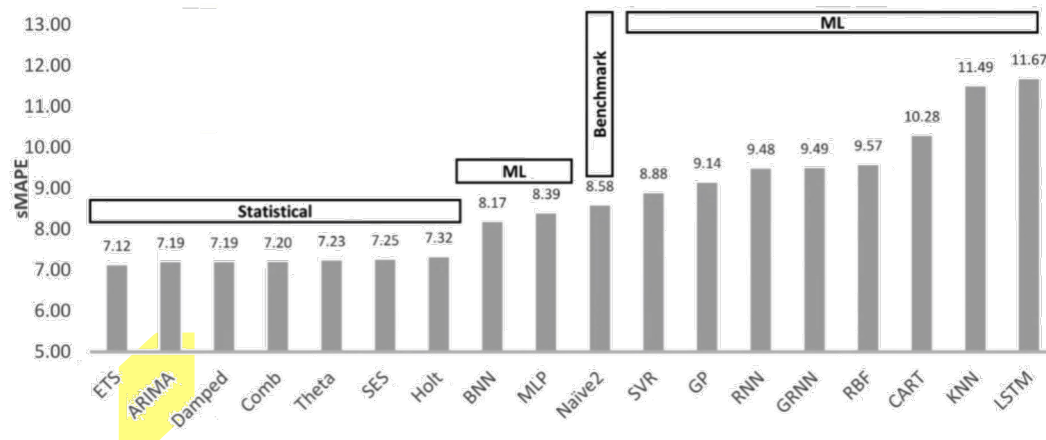
Many more advanced Deep Learning models

- D-Linear, N-Linear
- N-BEATS
- N-HiTS
- Temporal Convolutional Network
- Temporal Fusion Transformer
- TiDE - Time Series Dense Encoder
- TSMixer
- Space time former
- PatchTST
- TimesNet

Be aware of trade-offs!

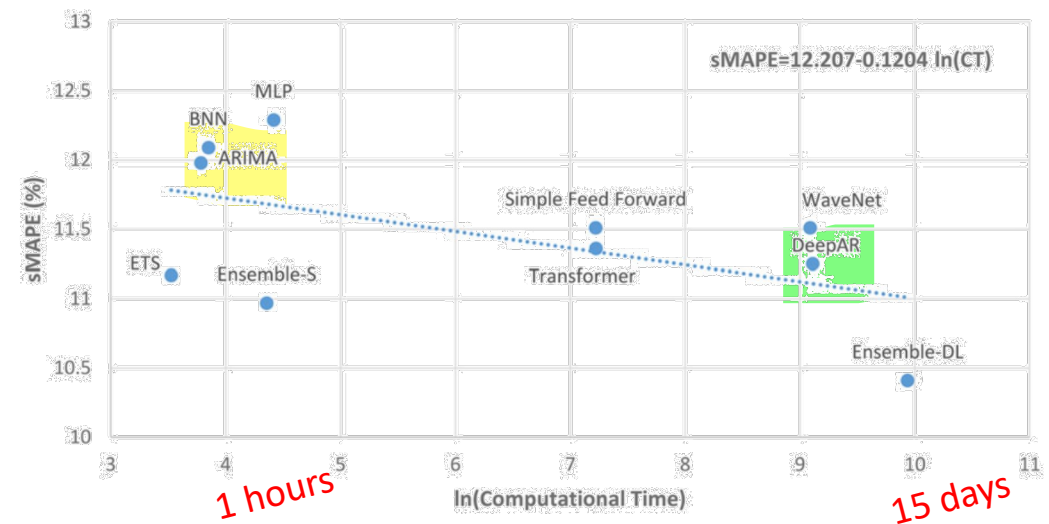
2018

Statistical methods largely outperform ML-based methods



2022

ML-based methods (i.e., deep learning ones) outperform statistical methods *but at a vast computational cost*



The advent of time series foundational models

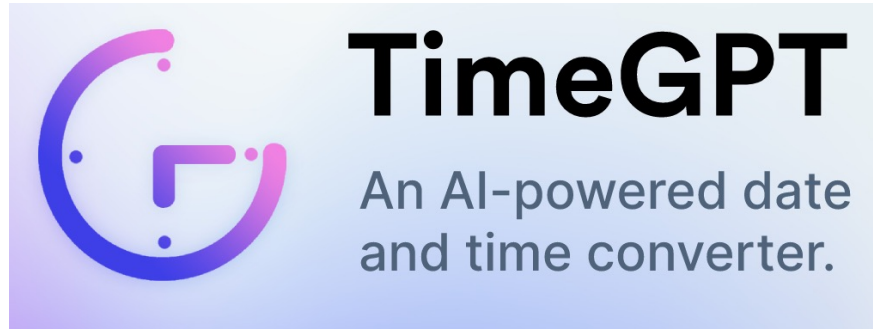
LLM

Task	Identify the years when a historical event happened.
In-context examples	Q: The French Revolution A: 1789 ~ 1799 Q: The first modern Olympic Games A: 1896 Q: Launch of the Voyager 2 A: 1977
Question	Q: Proof of Fermat's Last Theorem A: ?

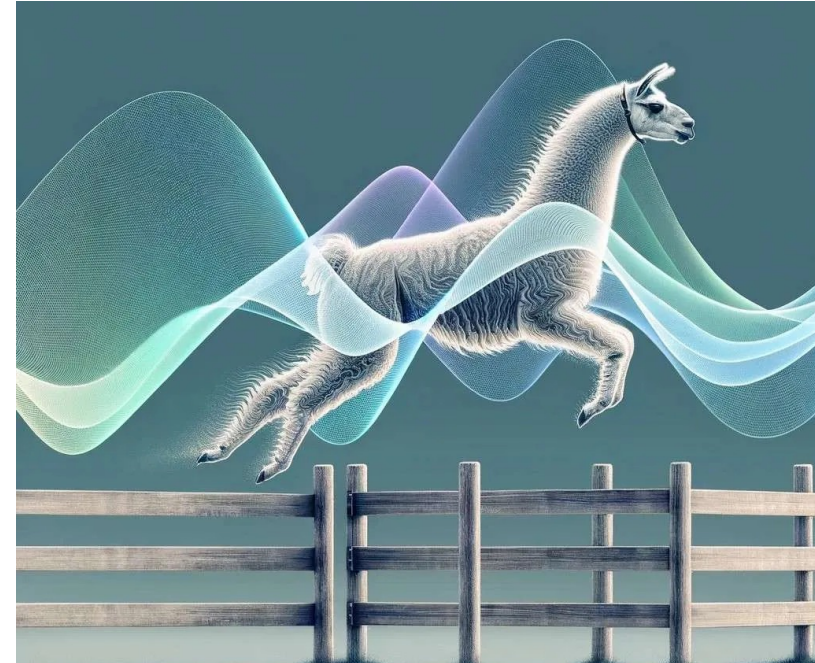
Time Series FM

Task	Forecast future hourly traffic on Highway 1.
In-context examples	Highway 2 Traffic Pattern Highway 3 Traffic Pattern Highway 4 Traffic Pattern
Question	Highway 1 Traffic 

The advent of time series foundational models



TimeGPT



Lag-Llama

<https://github.com/time-series-foundation-models/lag-llama>

<https://www.nixtla.io/timegpt>

More and more TS foundational models

- Time-LLM (Jin et al. 2023)
- TimeGPT-1 (Garza et al. 2023)
- Lag-Llama (Rasul et al. 2023)
- TimesFM (Das et al. 2023)
- Tiny Time Mixers (Ekambaram et al. 2024)
- Moirai (Woo et al. 2024)
- MOMENT (Goswami et al. 2024)
- UniTS (Gao et al. 2024)
- Chronos (Ansari et al. 2024)

<https://otexts.com/fpppy/nbs/15-foundation-models.html>

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